

- We can find ~~min~~ w_0, w_1 that give min $J(w_0, w_1)$ by taking partial derivative of $J(w)$ with respect to w_0, w_1 separately.

Approach Ordinary linear regression (OLS)

A closed form solution in terms of linear regression, refer to an analytical method to find the exact coefficient of polynomial by minimizing the error without iterative optimization.

$$\frac{\partial}{\partial w_0} J(w) = 0$$

$$\frac{\partial}{\partial w_1} J(w) = 0$$

$$\begin{cases} \hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x} = \frac{\sum_{i=1}^m y_i - \hat{w}_1 \sum_{i=1}^m x_i}{m} \\ \hat{w}_1 = \frac{\text{cov}(x, y)}{s_x^2} = r_{xy} \frac{s_y}{s_x} \end{cases}$$

→ Self-study for derivation

• There is a relation between slope and correlation coefficient

• There is a relation between slope and s_x and s_y .

Multiple linear regression

One dependent variable (y)

Multiple independent variable ($x_1, x_2, \dots, x_n$)

n = Dimension / feature

m = number of training samples.

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_n x_{in} + \epsilon_i$$

$$= \sum_{j=0}^n w_j x_{ij} + \epsilon_i \quad \left| \quad x_{i0} = 1 \right.$$

For hypothesis

$$\begin{aligned} \hat{y}_i = h(x_i) &= w_0 + w_1 x_{i0} + w_2 x_{i2} + \dots + w_n x_{in} \\ &= \sum_{j=0}^n w_j x_{ij} \quad | \quad x_{i0} = 1 \end{aligned}$$

$$x_i = [x_{i0} \quad x_{i1} \quad x_{i2} \quad \dots \quad x_{in}]$$

Same thing in matrix form:

$$Y = XW + \epsilon$$

m = training samples
 n = dimension

$$Y \rightarrow \overset{\text{dimension}}{m \times 1}$$

$$X \rightarrow m \times (n+1)$$

$$W \rightarrow (n+1) \times 1$$

$$\epsilon \rightarrow m \times 1$$

OLS estimation

$$W = (X^T X)^{-1} X^T Y$$

Observation (Self-study)

- useful for small dataset
- require more computing for large dataset
- Problem when $(X^T X)^{-1}$ is non-invertible.



- ① perfect multicollinearity
- ② more predictor than observer.

Coefficient of determination

$$R^2 = 1 - \frac{SS_{Res}}{SS_{tot}}$$

$$SS_{Res} = \text{Sum of squared residual} = \sum (y_i - \hat{y}_i)^2$$

$$SS_{tot} = \text{total sum of square} = \sum (y_i - \bar{y})^2$$

It is a statistical measure in linear regression that indicates how well the regression model explains the variability of dependent variable.

R^2 ranges 0 to 1

$R^2 = 1$ | The model is perfect fit; explain all the variance.

$R^2 = 0$ | The model is poor fit.

Exercise

Estimate y using x by OLS considering the following dataset.

x	y
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Estimate y for $x = 6$??

discussed in class